

# Bayesian reliability surveillance under sparse failures: distinguishing temporal anomalies from persistent degradation in a PSLV launch record

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## Abstract

Consecutive launch failures can be statistically unusual without proving that a launch system has permanently degraded. This study develops and demonstrates a three-layer Bayesian reliability-surveillance framework for sparse engineering data. The framework separates *surveillance* of unusual failure clustering, *diagnosis* of possible temporal reliability change, and *prediction* of future outcomes. We assemble a source-audited, frozen record of 64 chronological PSLV launches and compare a constant Beta–Bernoulli reliability model with an unknown change-point model, recent-window sensitivity analyses, smooth temporal trends, configuration-adjusted checks, posterior predictive diagnostics, and rolling-origin predictive validation. A 2,000-replicate-per-stream operating-characteristics study evaluates false alarms and limited detection power under constant, clustered, heterogeneous, abrupt-change, and smooth-change data-generating processes. The record contains 60 successes and 4 failures. Under the constant model, the posterior success probability is **Beta(61, 5)**, with posterior mean 0.9242 and 95% credible interval [0.8499, 0.9746]. The posterior predictive probability of a two-launch sequence containing two consecutive failures is 0.0068 under the constant model and 0.0312 under the change-point model. A decline is plausible, but the transition location is diffuse and the time-varying models do not improve held-out prediction: mean log scores are -0.1503, -0.1577, and -0.1507 for the constant, change-point, and smooth-trend models, respectively. The results support continued surveillance, but do not establish a persistent reliability change, a robust predictive advantage, or an engineering cause. The contribution is an auditable

evidence-synthesis workflow that prevents an unusual cluster, a possible temporal change, and a demonstrably better forecast from being treated as the same claim.

**Keywords:** Bayesian reliability; change-point analysis; launch reliability; posterior predictive checking; predictive validation; sparse failures; PSLV

## 1 Introduction

Reliability surveillance after clustered failures is difficult because the same observation can support several explanations. A pair of consecutive failures may indicate a genuine change in the underlying process, but it may also occur by chance under a high-reliability system observed for a finite number of launches. Treating the event as proof of degradation overstates the evidence; treating it as irrelevant ignores potentially important information. Bayesian inference is useful here because it reports uncertainty directly and supports sequential updating [1].

This paper studies that distinction using the Polar Satellite Launch Vehicle (PSLV) launch record. The primary research question is: *does the observed clustering of PSLV failures provide statistical evidence of a change in launch reliability relative to a constant-reliability baseline?* The question is deliberately narrower than whether PSLV is “safe” or “unsafe.” It concerns the statistical compatibility of the observed sequence with stable and time-varying reliability explanations.

The methodological contribution is an integrated and auditable surveillance workflow for sparse engineering reliability data. It combines source auditing, a constant Beta–Bernoulli model, sequential updating, unknown change-point inference, recent-window sensitivity, calendar-time and configuration checks, posterior predictive diagnostics, proper scoring rules, bootstrap uncertainty, and calibration diagnostics. The workflow is not presented as a new probability model. Its contribution is to make the evidential chain explicit and reproducible, while separating historical estimation, temporal diagnosis, and prediction of future launches. This distinction matters because a model can make a recent cluster look unusual without improving forecasts of subsequent outcomes.

The paper makes three distinctions. First, it asks whether the final two failures are unusual (surveillance). Second, it asks whether a decline in reliability is plausible, how large it might be, and how uncertain its timing is (diagnosis). Third, it asks whether a more complex time-varying model predicts future launches better (prediction). These questions need not have the same answer. In particular, an unusual cluster can produce posterior evidence of temporal heterogeneity without producing a reliable improvement in prospective forecasts.

The contributions are therefore threefold. First, we provide a reproducible surveillance–diagnosis–prediction decomposition for small-sample reliability records. Second, we combine exact clustering calculations, Bayesian posterior predictive quantities, interior and endpoint-sensitive temporal models, practical degradation probabilities, and proper-score validation in one evidence chain. Third, we demonstrate how the framework changes the interpretation of a real launch sequence: the PSLV record warrants continued surveillance, but the available evidence does not justify equating

recent clustering with a confirmed persistent degradation. The methodological contribution is an explicit decision architecture for separating evidence that a sequence is unusual, evidence that a reliability regime may have changed, and evidence that the change improves prospective forecasts.

Recent reliability research reinforces the need for this separation. Bayesian estimation is particularly useful when failure observations are scarce and prior assumptions can materially affect the result [2–4]. Modern reliability studies also use Bayesian updating to combine new evidence with existing engineering information, including reliability qualification testing, structural reliability updating, and traceable integration of sparse data with expert judgement [5–8]. Dynamic Bayesian networks and related probabilistic architectures are increasingly used for operational reliability, common-cause failures, multi-source evidence, and complex-system dependencies [9–15]. These models are valuable extensions when component-level or monitoring data are available, but they should not be imposed on a short binary launch sequence without sufficient information. Recent work on Bayesian change-point analysis and degradation modelling likewise emphasizes uncertainty in transition locations, heterogeneity, and the distinction between detecting a regime change and predicting future performance [16–18]. Finally, predictive-maintenance studies demonstrate the importance of prospective validation and decision relevance rather than relying only on in-sample fit [19–22]. The present study occupies a complementary position: it offers a deliberately transparent, low-dimensional surveillance framework for a source-audited sequence in which the number of failures is too small to support a fully parameterized dynamic system model.

## 2 Data and source verification

The analysis uses a frozen Version 1.0 dataset containing 64 chronological PSLV launches. The dataset has 60 successful and 4 unsuccessful primary mission outcomes, corresponding to an observed success proportion of 0.9375. The unsuccessful missions are PSLV-D1, PSLV-C39, PSLV-C61, and PSLV-C62. The chronology and mission outcomes are based on official Indian Space Research Organisation records [23, 24]. Each row retains the launch date, mission identifier, source-facing outcome text, outcome verification source, source disagreement indicator, launch configuration, failure-stage information, cause-status information, and a previous-failure indicator.

The dataset was constructed before inferential modeling. A final audit records 64 launches, zero duplicate missions, zero missing dates, zero missing outcomes, and zero unverified outcomes. One source conflict, two missing or unknown configuration values, and five missing payload fields are recorded as reviewed metadata limitations rather than silently imputed values. The C62 outcome is resolved using the dedicated official mission catalogue because the launch-list remark is blank. The data-freeze rule is that any future correction creates a new dataset version rather than silently changing the analysis input.

The statistical unit is one launch. A success is coded as  $Y_i = 1$  and an unsuccessful primary mission as  $Y_i = 0$ . Failure-stage descriptions are treated as descriptive observations, not as proven root causes. In particular, the statistical analysis does not establish that a third-stage anomaly caused a change in overall vehicle reliability.

### 3 Statistical framework

#### 3.1 Estimands and methodological principles

The first estimand is the overall success probability

$$p = \Pr(Y_i = 1). \quad (1)$$

The second is the possible reliability difference under a two-regime model,

$$\Delta = p_{\text{before}} - p_{\text{after}}, \quad (2)$$

with practical quantities  $\Pr(\Delta > 0)$ ,  $\Pr(\Delta > 0.05)$ , and  $\Pr(\Delta > 0.10)$ . The third estimand is predictive performance, evaluated using log score, Brier score, and calibration on held-out launches.

We use two methodological principles to organize interpretation. *Principle 1 (surveillance–diagnosis)*: an unusual failure cluster can generate evidence of temporal heterogeneity without identifying persistent reliability degradation. *Principle 2 (diagnosis–prediction)*: evidence of temporal heterogeneity should not be treated as practically useful predictive structure unless it improves prospective forecasting relative to the constant-reliability baseline. These are reporting principles, not mathematical propositions or claims that the PSLV record must support a decline. Their operating characteristics are examined in simulation, and their predictive implication is evaluated on future observations.

#### 3.2 Surveillance–diagnosis–prediction decomposition

The analysis is organized into three layers. The surveillance layer asks whether the observed ordering of failures is unusual under a stable reference process. It uses the conditional runs probability, posterior predictive probabilities, prior predictive checks, and failure-gap summaries. The diagnosis layer asks whether the reliability parameter may differ over time. It uses the unknown change-point model, posterior uncertainty in the transition location and degradation magnitude, smooth temporal trends, and configuration-adjusted sensitivities. The prediction layer asks whether the additional temporal structure improves forecasts for launches that were not used for fitting. It uses rolling-origin log scores, Brier scores, calibration, and bootstrap uncertainty.

This decomposition prevents a common interpretive error: a small probability for a particular observed cluster is evidence that the cluster is unusual under a reference model, but it is not by itself evidence that a persistent engineering degradation exists. Likewise, a posterior probability such as  $\Pr(\Delta > 0)$  measures uncertainty about a model-defined contrast; it does not establish causal failure attribution.

The constant model serves as a reference model:

$$H_0 : p_1 = p_2 = \cdots = p_{64}. \quad (3)$$

**Table 1** Prior sensitivity for the constant-reliability model.

| Prior          | Posterior mean | 95% CrI          | Pr(two failures) |
|----------------|----------------|------------------|------------------|
| Beta(0.5, 0.5) | 0.9308         | [0.8582, 0.9785] | 0.00577          |
| Beta(1, 1)     | 0.9242         | [0.8499, 0.9746] | 0.00678          |
| Beta(2, 2)     | 0.9118         | [0.8344, 0.9664] | 0.00895          |
| Beta(5, 1)     | 0.9286         | [0.8582, 0.9761] | 0.00604          |
| Beta(9, 1)     | 0.9324         | [0.8656, 0.9774] | 0.00541          |

The competing change-point model allows

$$p_i = \begin{cases} p_1, & i \leq k, \\ p_2, & i > k, \end{cases} \quad p_1 \neq p_2. \quad (4)$$

These are Bayesian model comparisons rather than classical null-hypothesis tests. The constant model is not interpreted as a claim that reliability can never change.

### 3.3 Constant-reliability baseline

Under  $M_0$ ,  $Y_i \mid p \sim \text{Bernoulli}(p)$  and  $p \sim \text{Beta}(1, 1)$ . With  $S = 60$  successes and  $F = 4$  failures,

$$p \mid D \sim \text{Beta}(1 + S, 1 + F) = \text{Beta}(61, 5). \quad (5)$$

The posterior mean is 0.9242 and the central 95% credible interval is [0.8499, 0.9746]. Prior sensitivity uses Jeffreys Beta(0.5, 0.5), Beta(2, 2), and reliability-oriented alternatives. Table 1 reports the corresponding posterior summaries and the posterior predictive probability of two consecutive failures.

### 3.4 Temporal models and sensitivity analyses

The primary unknown change-point model  $M_1$  is an *interior-change model*: it evaluates candidate locations  $k \in \{10, \dots, 54\}$ , leaving at least ten launches in each segment. Independent Beta(1, 1) priors are assigned to the segment reliabilities. The marginal likelihood integrates each segment probability analytically, following the general Bayesian change-point perspective of [25]. This minimum-segment restriction is deliberate: it prevents a one- or two-observation segment from dominating a two-regime comparison, but it means that  $M_1$  is not designed to represent an abrupt change at launches 63–64.

The endpoint-sensitive model  $M_2$  complements, rather than replaces,  $M_1$ . It compares the historical launches with the last 2, 3, 4, 5, and 6 launches under an explicitly stated set of recent-window hypotheses. All five windows are reported; no single window is treated as a pre-specified change point. The two-launch window is especially sensitive because it is motivated by the observed C61/C62 pair and contains only two observations, so its Bayes factor is a boundary-sensitivity result, not confirmatory evidence. The smooth trend model  $M_3$  uses a Bayesian logistic trend with a Normal(0, 2.5) intercept prior and Normal(0, 1) slope prior evaluated on a numerical grid. Both launch-index time and normalized calendar time are examined.

Configuration-adjusted analyses use penalized logistic fits with configuration indicators and, separately, with launch-index or calendar time. These are descriptive sensitivity models, not primary inferential models, because only four failures are observed and several configuration groups are sparse.

### 3.5 Operating-characteristics simulation

To assess how the surveillance–diagnosis layer behaves under sparse data, we conducted an operating-characteristics simulation using the same  $n = 64$  launch length, Beta–Bernoulli baseline, and interior  $k = 10, \dots, 54$  change-point grid as the empirical  $M_1$  analysis. Six labelled streams were generated: a constant-reliability reference ( $p = 0.94$ ), an independently seeded repeat of that same constant process to provide a Monte Carlo replicate, an abrupt decline from 0.97 to 0.80 after launch 32, a mild abrupt decline from 0.96 to 0.91, a smooth decline from 0.97 to 0.85, and alternating configuration-specific probabilities (0.98, 0.94, 0.90, 0.96) without a temporal trend. The first two streams are therefore not different scientific mechanisms; they check stochastic variability under the same constant Bernoulli reference. For each labelled stream, 2,000 datasets were generated with seed 20261002.

For every simulated dataset, we recomputed the posterior change-point distribution and estimated  $\Pr(\Delta > 0)$  and  $\Pr(\Delta > 0.05)$ . A secondary practical-decline flag was recorded when  $\Pr(\Delta > 0.05) > 0.95$ . This 0.95 value is a deliberately stringent reporting threshold, not a calibrated operational decision limit; the continuous posterior probabilities remain the primary output. Wilson 95% intervals quantify Monte Carlo uncertainty in each flag rate. The simulation is an operating-characteristics check, not a claim that any scenario represents PSLV. In particular, the results quantify both false-positive behavior under stable or heterogeneous data and the limited detection power that is unavoidable with only 64 observations.

### 3.6 Predictive validation

The rolling-origin design is a historical-stability benchmark: its last scored block is launches 51–60 and therefore does not score C61 or C62. The one-step-ahead prequential design below supplements it by scoring the final launches after their preceding histories.

Models are evaluated at training cutoffs 20, 25, 30, 35, 40, 45, and 50, with the following ten launches held out at each cutoff. Proper scoring rules are used because predictive evaluation should reward calibrated probabilistic forecasts [26, 27]. For predicted success probabilities  $q_i$ , the mean log score and Brier score are

$$\text{LS} = \frac{1}{m} \sum_{i=1}^m [y_i \log q_i + (1 - y_i) \log(1 - q_i)], \quad \text{BS} = \frac{1}{m} \sum_{i=1}^m (q_i - y_i)^2. \quad (6)$$

Higher log scores and lower Brier scores are preferable. The seven paired windows are resampled 10,000 times to quantify uncertainty in the M1–M0 score difference using a bootstrap procedure [28]. Because the windows overlap and contain few failures, this bootstrap is treated as a descriptive uncertainty assessment rather than a large-sample

inferential guarantee. Calibration is reported as an exploratory comparison of mean predicted probability and observed success frequency in held-out probability bins.

For interpretation, the predictive value of a temporal model is summarized by

$$\Delta_{\text{pred}} = \text{LS}_{\text{complex}} - \text{LS}_{\text{baseline}}. \quad (7)$$

A positive value indicates better average predictive log score for the complex model, whereas a value near zero indicates that the additional structure has not demonstrated practical forecasting value. This predictive contrast is kept separate from the posterior degradation contrast  $\Delta$ .

As a complementary surveillance-oriented check, we also perform one-step-ahead prequential prediction. For each training size from 20 through 63, the next launch is predicted using only the preceding outcomes. The constant model uses the Beta posterior predictive mean. The change-point model averages segment-specific posterior predictive probabilities over its marginal posterior for  $k \in \{10, \dots, n - 10\}$ , where  $n$  is the current training size. Scores are accumulated over the resulting 44 sequential predictions. This analysis is reported separately from the seven overlapping ten-launch test blocks because the two validation designs answer different questions.

## 4 Results

### 4.1 Operating characteristics under sparse data

The simulation shows that the practical-decline rule is conservative under stable or non-temporal alternatives. The practical-decline flag rate was 0.50% (95% Monte Carlo interval [0.27%,0.92%]) for the constant reference, 0.25% ([0.11%,0.58%]) for its independent replicate, and 0.40% ([0.20%,0.79%]) under configuration heterogeneity. Detection increased to 1.85% ([1.35%,2.54%]) for the mild abrupt decline, 10.30% ([9.04%,11.71%]) for the smooth decline, and 28.75% ([26.81%,30.77%]) for the strong abrupt decline. Mean posterior probabilities of any decline were 0.505, 0.495, 0.504, 0.678, 0.813, and 0.900 for the constant, replicate-constant, configuration-heterogeneity, mild-abrupt, smooth, and strong-abrupt streams, respectively.

These results provide two important qualifications. First, the observed flag rates under stable or heterogeneous streams were only 0.25–0.50%, but a low observed rate is not evidence that the true false-positive probability is exactly zero. Second, even a strong abrupt change was flagged in only 28.75% of samples of length 64. The simulation therefore supports the framework’s intended interpretation: sparse data can control false alarms reasonably well, but they cannot provide high-power detection of every plausible degradation pattern.

Across the five prior choices, posterior means range from 0.9118 to 0.9324 and all 95% credible intervals overlap substantially. The posterior predictive probability of two consecutive failures ranges from 0.00541 to 0.00895. The qualitative conclusion is therefore stable under the examined prior choices: the cluster is unusual under a high-reliability reference, but prior variation does not establish persistent degradation.

**Table 2** Operating characteristics of the change-point layer in 2,000 simulated datasets per labelled stream. The practical-decline flag is  $\Pr(\Delta > 0.05) > 0.95$ ; intervals are Wilson 95% Monte Carlo intervals. The repeat-constant stream uses the same constant Bernoulli mechanism as the constant reference with an independent random stream.

| Scenario                              | $\Pr(\Delta > 0)$<br>mean | Practical<br>flag rate | Mean<br>mode $k$ | Mean posterior<br>success |
|---------------------------------------|---------------------------|------------------------|------------------|---------------------------|
| Constant reference ( $p = 0.94$ )     | 0.505                     | 0.005                  | 30.32            | 0.927                     |
| Repeat constant stream ( $p = 0.94$ ) | 0.495                     | 0.003                  | 29.68            | 0.927                     |
| Configuration heterogeneity           | 0.504                     | 0.004                  | 30.72            | 0.931                     |
| Mild abrupt decline                   | 0.678                     | 0.019                  | 35.78            | 0.922                     |
| Smooth decline                        | 0.813                     | 0.103                  | 38.28            | 0.897                     |
| Strong abrupt decline                 | 0.900                     | 0.288                  | 37.24            | 0.873                     |

## 4.2 Historical reliability and clustered failures

The observed success proportion is 93.75% and the Wilson 95% interval is [0.8500, 0.9754]. Failures occur at launch indices 1, 41, 63, and 64, with the final two failures consecutive. The exact lower-tail runs probability is 0.0090 under the observed success and failure margins, indicating that this ordering is unusual under that conditional reference. This result motivates temporal analysis but does not identify a mechanism or prove a permanent change.

Under  $M_0$ , the exact posterior predictive probability that the next two launches are both failures is

$$\mathbb{E}[(1-p)^2 \mid D] = \frac{5 \cdot 6}{66 \cdot 67} = 0.0068. \quad (8)$$

Under  $M_1$ , averaging over posterior uncertainty in  $k$ , the corresponding posterior predictive probability is 0.0312. Thus, allowing temporal heterogeneity makes the pair less surprising, but the event remains uncommon even under the change-point model.

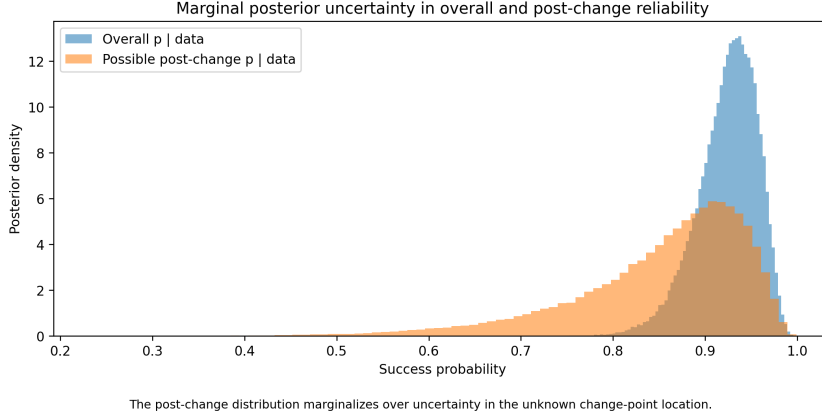
## 4.3 Temporal evidence

The interior  $M_1$  posterior has mode  $k = 54$ , mean 37.42, and central 95% interval [11, 54]. Because 54 is the upper boundary of the interior grid, the mode must not be interpreted as an identified transition at launch 54; it indicates that the interior model places substantial mass toward its recent boundary. The posterior probabilities are  $\Pr(\Delta > 0) = 0.7824$ ,  $\Pr(\Delta > 0.05) = 0.5795$ , and  $\Pr(\Delta > 0.10) = 0.3746$ . These values support a possible decline but do not provide decisive evidence for a practically large degradation. The broad interval for  $k$  is more important than the mode alone: the interior model does not identify a unique transition.

The endpoint-sensitive recent-window analysis gives  $\Pr(p_{\text{old}} > p_{\text{recent}}) = 0.9998$  for the last two launches and 0.9931 for the last six. This persistence across nearby windows is useful for surveillance, but it is not confirmatory because the window family

was examined after the failure sequence was observed and the number of failures is small. The Bayes factors relative to  $M_0$  are 0.300 for the interior  $M_1$ , 115.56 for the two-launch member of the exploratory  $M_2$  window family at  $k = 62$ , and 1.320 for  $M_3$ . The  $M_2$  comparison answers a different question from the interior change-point model and is not interchangeable with it; in particular, the large two-launch value should not be reported as evidence for a uniquely identified change point.

Launch-index time gives posterior probability 0.7747 for a negative smooth trend, whereas calendar time gives 0.5492. This difference shows that temporal evidence is sensitive to the representation of time. Configuration-adjusted fits are similarly limited by sparse failures and are retained as robustness checks rather than headline evidence.



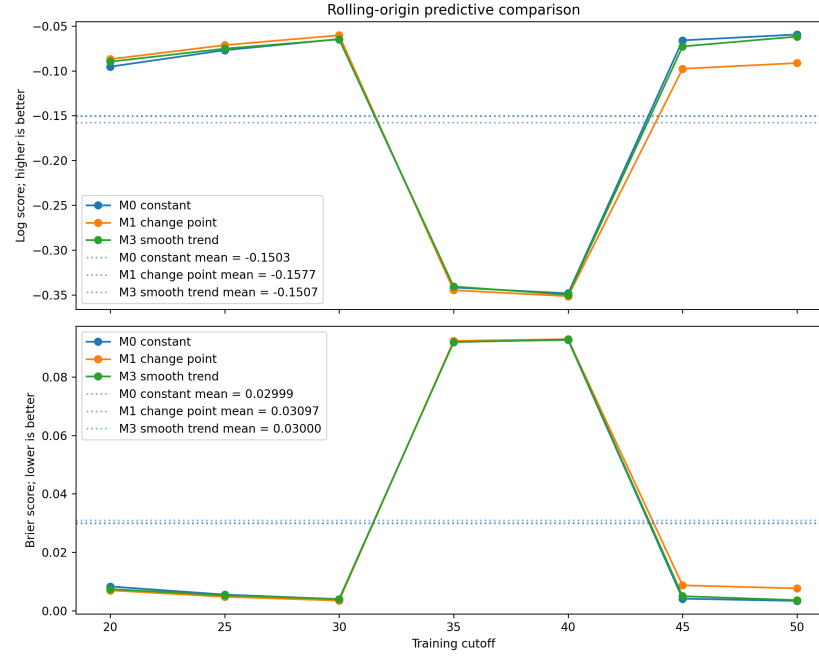
**Fig. 1** Marginal posterior uncertainty in overall and possible post-change reliability. The post-change distribution marginalizes over uncertainty in the unknown change-point location.

#### 4.4 Predictive validation

Mean log scores over the seven historical rolling-origin windows are -0.1503 for  $M_0$ , -0.1577 for  $M_1$ , and -0.1507 for  $M_3$ . Mean Brier scores are 0.02999, 0.03097, and 0.03000, respectively. Hence the constant model is slightly better on average, while the differences are small. The bootstrap 95% interval for the  $M_1$ – $M_0$  log-score difference is [-0.0195, 0.0035], which includes zero. Because these blocks stop before C61 and C62, this result is evidence about historical forecast stability, not a direct test of the final failure pair. The predictive results do not support replacing the simple reference model with a more complex model for forecasting on this dataset.

The complementary one-step-ahead analysis scores 44 sequential predictions, including the final launches after their preceding histories. In this design,  $M_1$  has a mean log score of -0.2592 and mean Brier score of 0.06463, compared with -0.2613 and 0.06491 for  $M_0$ . The paired bootstrap log-score difference  $M_1 - M_0$  is 0.002105 with a 95% interval of [-0.012980, 0.025144]; the paired Brier difference is -0.000273 with a 95% interval of [-0.003411, 0.001707]. Thus, the change-point model is marginally better in the strictly sequential design, but both intervals include zero. The two validation designs therefore

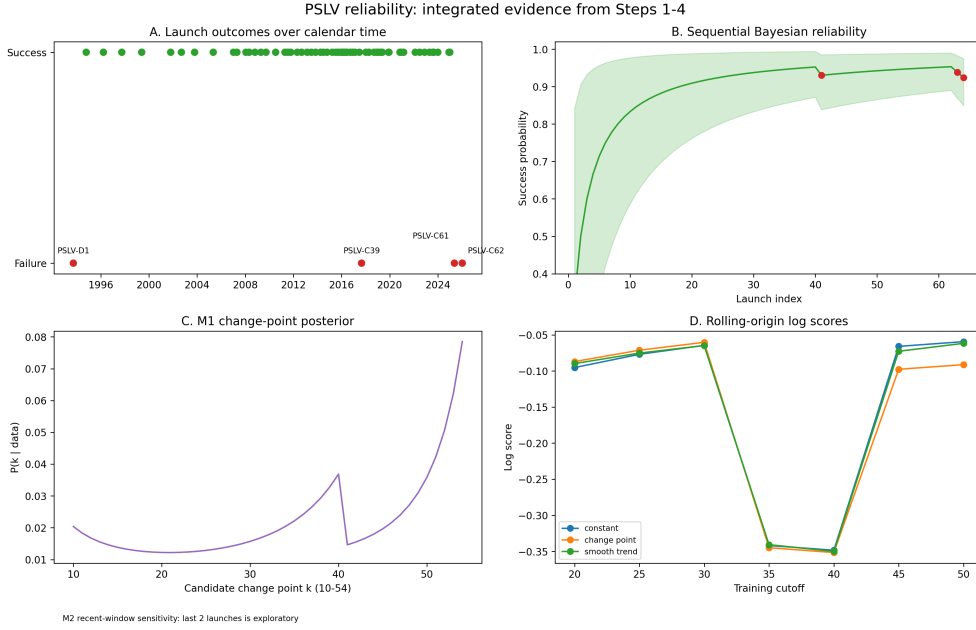
lead to the same cautious decision: temporal structure is worth monitoring, including at the endpoint, but model complexity is not yet justified as the default forecasting model.



**Fig. 2** Rolling-origin predictive comparison across seven ten-launch test blocks. Higher log scores are better and lower Brier scores are better. Dotted lines show model means across the seven paired windows.

**Table 3** Summary of statistical evidence.

| Quantity                                                            | Estimate or result              |
|---------------------------------------------------------------------|---------------------------------|
| Launches                                                            | 64                              |
| Successes / failures                                                | 60 / 4                          |
| Observed success rate                                               | 0.9375                          |
| Wilson 95% interval                                                 | [0.8500, 0.9754]                |
| $M_0$ posterior mean and 95% CrI                                    | 0.9242; [0.8499, 0.9746]        |
| Posterior predictive probability of two consecutive failures, $M_0$ | 0.0068                          |
| Posterior predictive probability of two consecutive failures, $M_1$ | 0.0312                          |
| $M_1$ interior-grid mode / mean of $k$                              | 54 / 37.42                      |
| $M_1$ 95% interval for $k$                                          | [11, 54]                        |
| $\Pr(\Delta > 0)$ , $\Pr(\Delta > 0.05)$ , $\Pr(\Delta > 0.10)$     | 0.7824; 0.5795; 0.3746          |
| $\Pr(p_{old} > p_{recent})$ , last 2 / last 6                       | 0.9998 / 0.9931                 |
| Bayes factors $M_1/M_0$ , $M_2/M_0$ , $M_3/M_0$                     | 0.300; 115.56; 1.320            |
| Mean log scores $M_0$ , $M_1$ , $M_3$                               | -0.1503; -0.1577; -0.1507       |
| Bootstrap 95% interval for $M_1-M_0$ log-score difference           | [-0.0195, 0.0035]               |
| Sequential one-step log scores, $M_0$ / $M_1$                       | -0.2613 / -0.2592               |
| Sequential one-step Brier scores, $M_0$ / $M_1$                     | 0.06491 / 0.06463               |
| Sequential log-score difference, $M_1 - M_0$ (95% interval)         | 0.002105 [-0.012980, 0.025144]  |
| Sequential Brier difference, $M_1 - M_0$ (95% interval)             | -0.000273 [-0.003411, 0.001707] |



**Fig. 3** Integrated evidence from the PSLV reliability analysis. Panel A shows the calendar-time outcome chronology, with unsuccessful missions annotated. Panel B shows sequential Bayesian updating under the constant model. Panel C shows the interior  $M_1$  posterior over candidate change points  $k = 10, \dots, 54$ ; endpoint-sensitive recent windows are analyzed separately and are not plotted as  $M_1$  candidates. Panel D shows historical rolling-origin log scores.

## 5 Engineering interpretation and causal boundary

The statistical record should be interpreted alongside, but not substituted for, engineering investigation. The final two unsuccessful missions are temporally adjacent and therefore motivate a focused reliability review. The statistical framework can quantify how unusual their ordering is, estimate uncertainty about possible temporal change, and test whether temporal structure improves prediction. It cannot identify a failed component, establish a common physical mechanism, or prove that an observed stage-level anomaly caused a long-term change in vehicle reliability.

This distinction separates three claims that are often conflated. A *statistical claim* concerns compatibility of the launch sequence with a probability model. An *engineering claim* concerns the observed failure stage or subsystem. A *causal claim* concerns whether that subsystem produced a persistent reliability change. The present study addresses the first claim and uses the second only as descriptive metadata; it does not establish the third. This boundary is especially important when the number of failures is small and several launches may share design, manufacturing, or operational conditions.

## 6 Discussion

The evidence is best understood through the proposed surveillance–diagnosis–prediction decomposition. At the surveillance layer, the final pair is unusual under a constant model. At the diagnosis layer, the interior temporal model makes a decline plausible but pushes mass toward its boundary, while the endpoint-sensitive analysis shows that the final pair is highly influential. Neither analysis identifies a unique transition or separates a persistent change from a short-lived cluster. Recent-window probabilities are strong but exploratory because the window family was examined after observing the failures. At the prediction layer, the historical rolling-origin design slightly favors the constant model, while the endpoint-inclusive sequential design gives the change-point model a numerically small advantage whose uncertainty interval includes zero. Taken together, the results argue for monitoring and data collection rather than a categorical declaration of permanent degradation.

This pattern has a practical implication. Statistical surveillance should continue, but the present record does not justify a categorical statement that PSLV reliability has permanently declined. Under the constant model, a future success updates the posterior from  $\text{Beta}(61, 5)$  to  $\text{Beta}(62, 5)$ , while a future failure updates it to  $\text{Beta}(61, 6)$ . These updates are specific to  $M_0$  and do not replace re-fitting the temporal models when new data arrive.

The framework also provides a disciplined way to report practical importance. A success probability of 0.925 corresponds to approximately 7.5 hypothetical failures per 100 launches, while probabilities of 0.95, 0.90, 0.85, and 0.80 correspond to 5, 10, 15, and 20 hypothetical failures per 100 launches. These translations are communication aids, not engineering safety thresholds.

## 7 Limitations

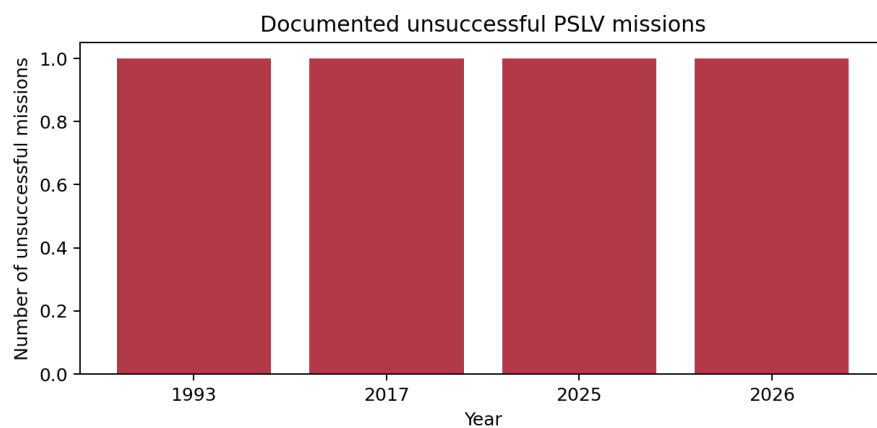
Only four failures are observed. This is sufficient to describe historical reliability but limits inference about temporal and configuration-specific effects. Conditional independence and exchangeability may be imperfect if launches share operational, manufacturing, or mission conditions. The configuration groups are sparse, and the recent two-launch window is especially fragile. Calendar time and launch index are both defensible but answer different temporal questions. The interior change-point model deliberately avoids tiny terminal segments, while the endpoint-sensitive model has very little data by construction; their roles should not be conflated. The data support statistical association with time, not causal attribution to a failure stage, configuration, or engineering mechanism. The seven-block validation uses overlapping test windows and stops before C61/C62, while the endpoint-inclusive sequential analysis contains only 44 one-step-ahead predictions; neither design provides a large-sample guarantee. The framework should therefore be viewed as a surveillance and evidence-organization tool, not as a substitute for engineering failure investigation or a controlled reliability demonstration test.

## 8 Conclusion

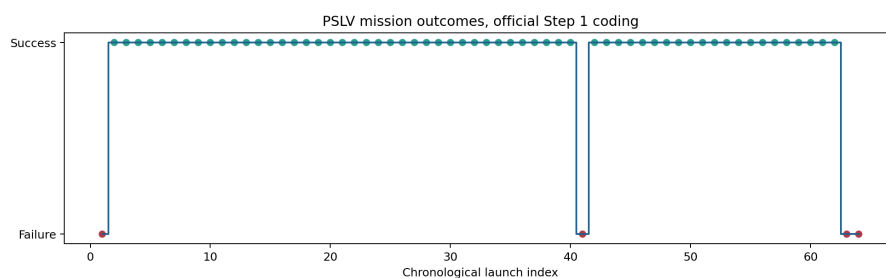
The audited PSLV record contains 60 successes and 4 failures in 64 launches. The final two failures are unusual under a constant-reliability model and justify continued surveillance. A recent reliability decline is compatible with the data, but its magnitude and timing remain uncertain, and calendar-time sensitivity weakens the smooth-trend evidence. The seven-block validation slightly favors the constant model, whereas the sequential one-step-ahead analysis gives a small advantage to the change-point model; neither design establishes a robust predictive improvement. The defensible conclusion is therefore that recent degradation is plausible but not conclusively established as a persistent reliability change. The broader contribution is a reproducible Bayesian workflow for making that distinction explicit in small-sample engineering reliability studies and for preventing exploratory temporal signals from being confused with validated predictive improvement.

## A Supplementary figures

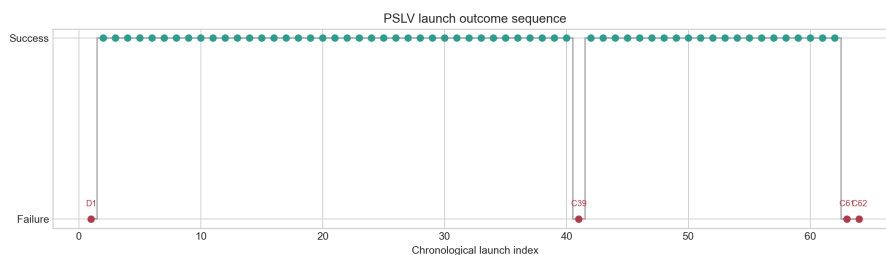
The reproducibility package contains the complete diagnostic archive. The following supplementary panels preserve the analyses used to support the synthesis; the main text reports only the figures needed for the central statistical argument.



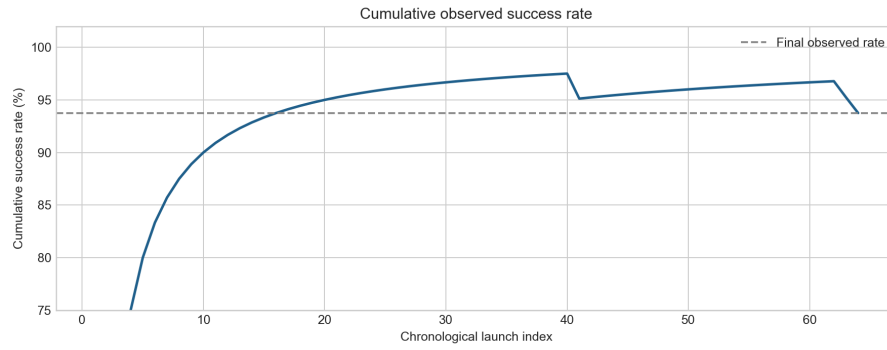
**Fig. 4** Supplementary diagnostic plot from the reproducibility archive (source identifier: failures\_by\_year).



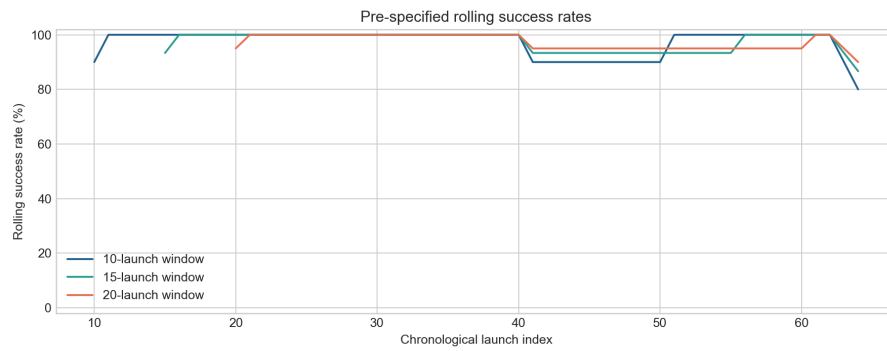
**Fig. 5** Supplementary diagnostic plot from the reproducibility archive (source identifier: success\_sequence).



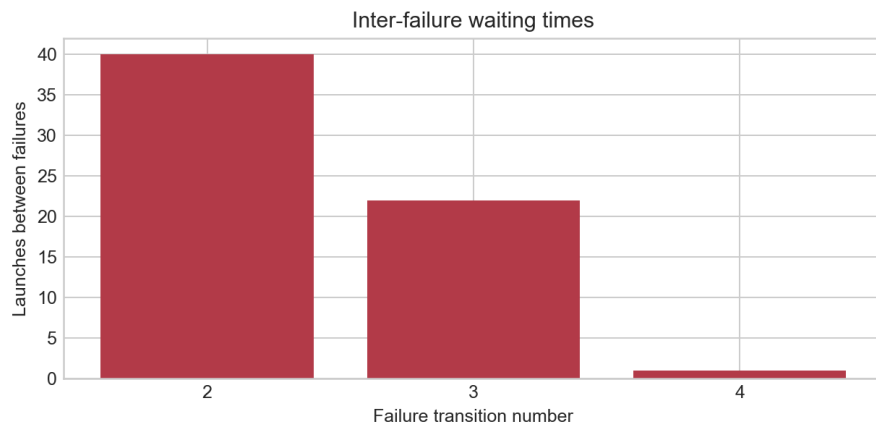
**Fig. 6** Supplementary diagnostic plot from the reproducibility archive (source identifier: step2\_01\_outcome\_sequence).



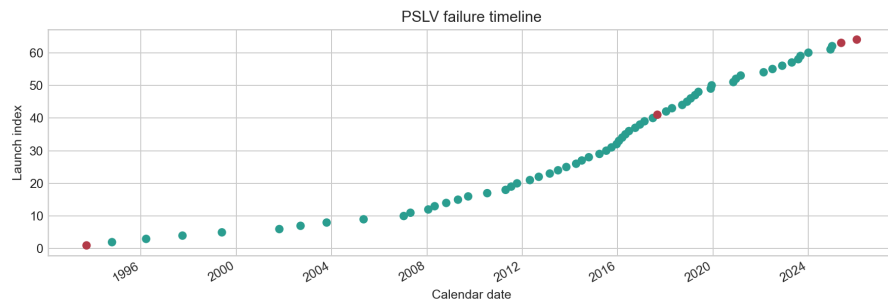
**Fig. 7** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step2_02_cumulative_success`).



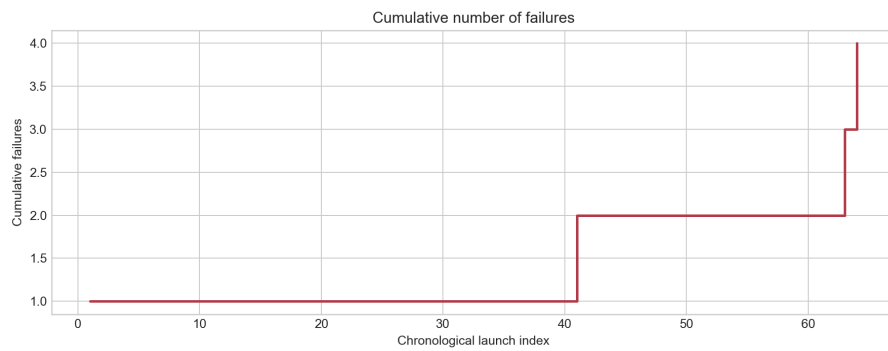
**Fig. 8** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step2_03_rolling_success`).



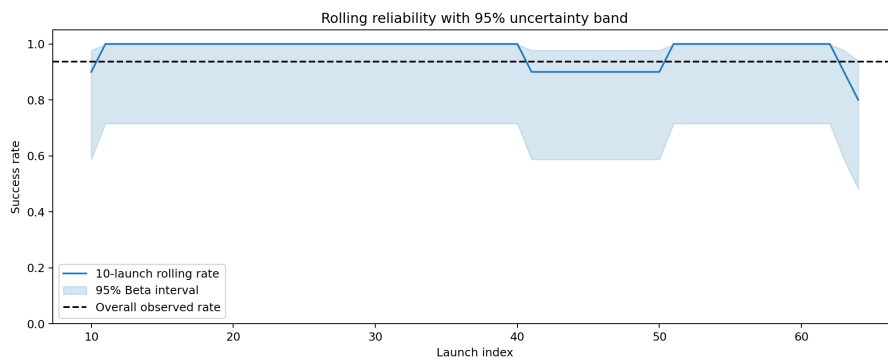
**Fig. 9** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step2_04_failure_gaps`).



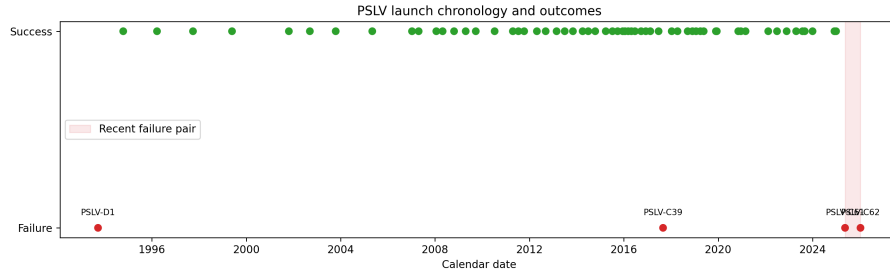
**Fig. 10** Supplementary diagnostic plot from the reproducibility archive (source identifier: step2\_05\_failure\_timeline).



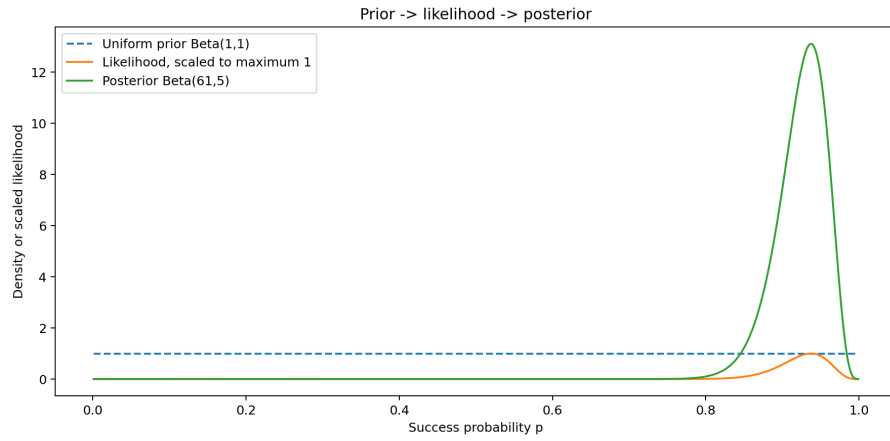
**Fig. 11** Supplementary diagnostic plot from the reproducibility archive (source identifier: step2\_06\_cumulative\_failures).



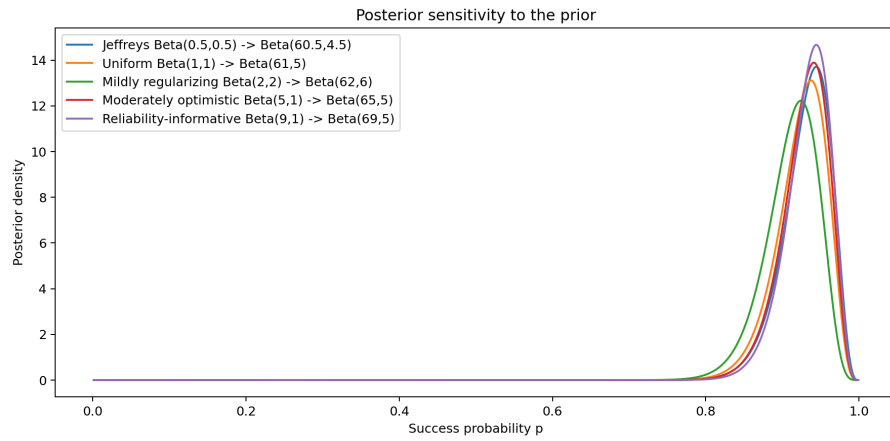
**Fig. 12** Supplementary diagnostic plot from the reproducibility archive (source identifier: step2\_07\_rolling\_uncertainty).



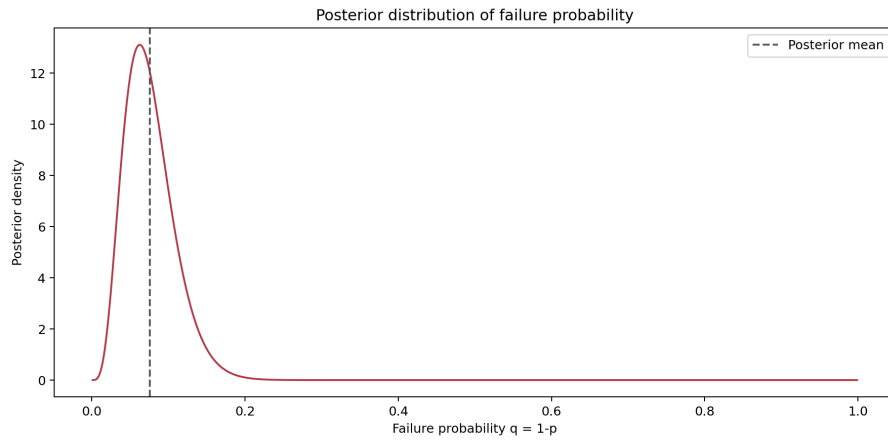
**Fig. 13** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step2_08_failure_chronology_publication`).



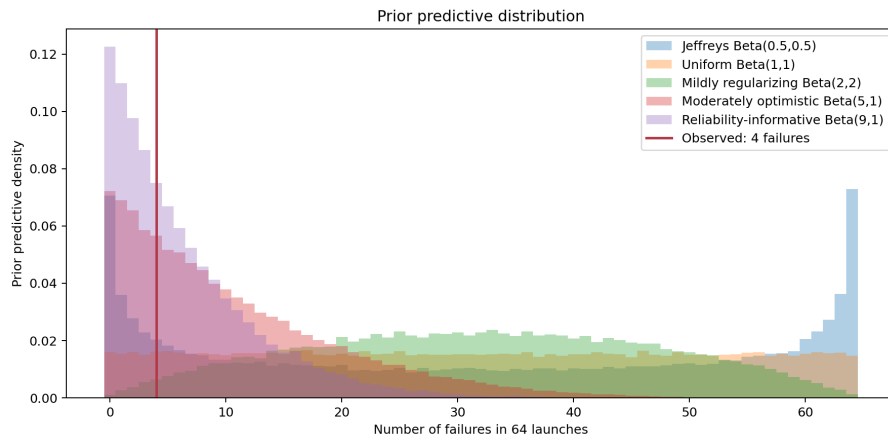
**Fig. 14** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step3_01_prior_likelihood_posterior`).



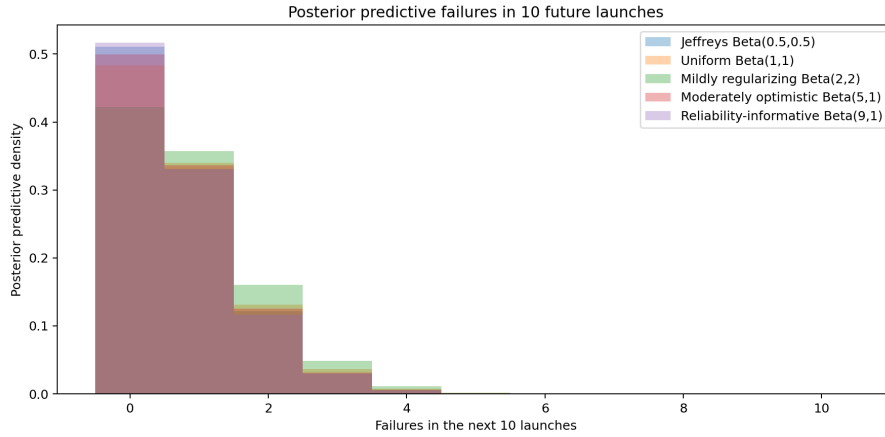
**Fig. 15** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step3_02_prior_sensitivity`).



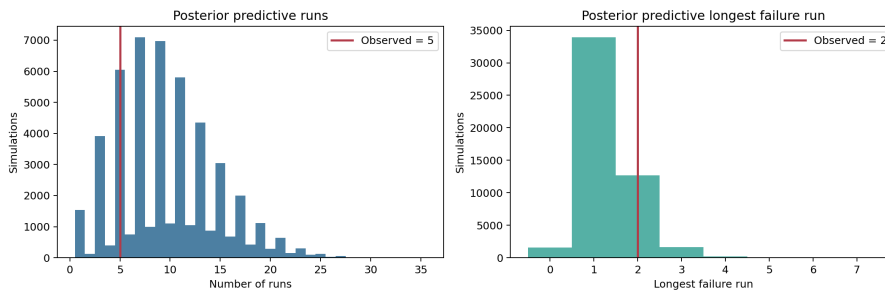
**Fig. 16** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step3_03_failure_probability`).



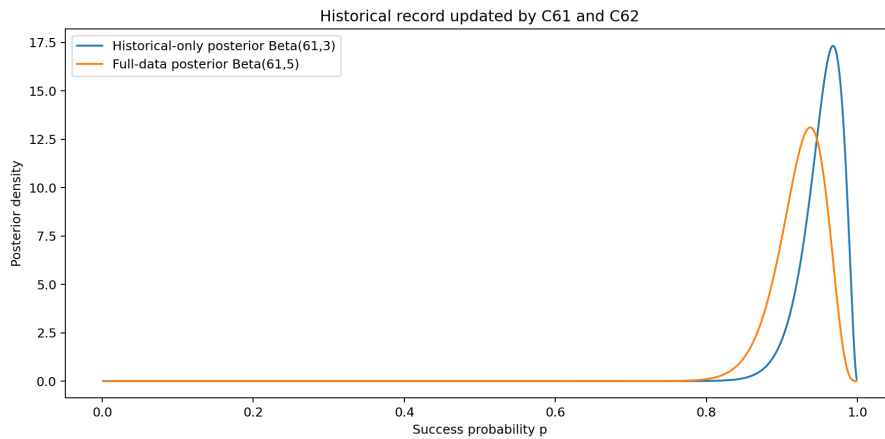
**Fig. 17** Supplementary diagnostic plot from the reproducibility archive (source identifier: `step3_04_prior_predictive`).



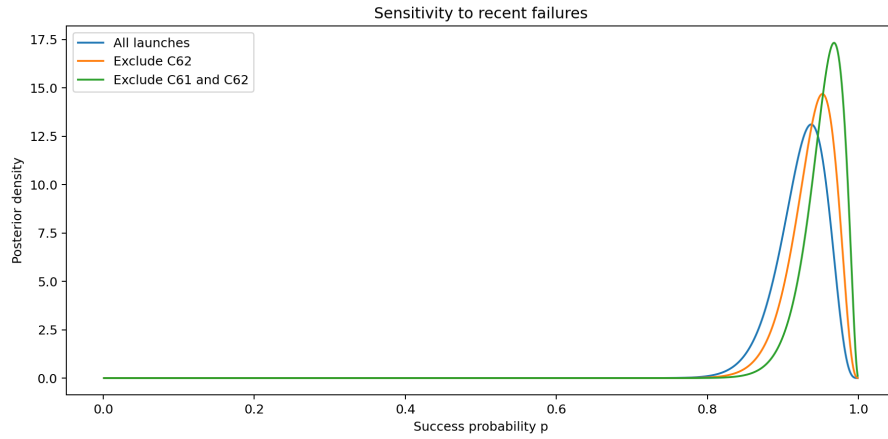
**Fig. 18** Supplementary diagnostic plot from the reproducibility archive (source identifier: step3\_05\_posterior\_predictive\_10).



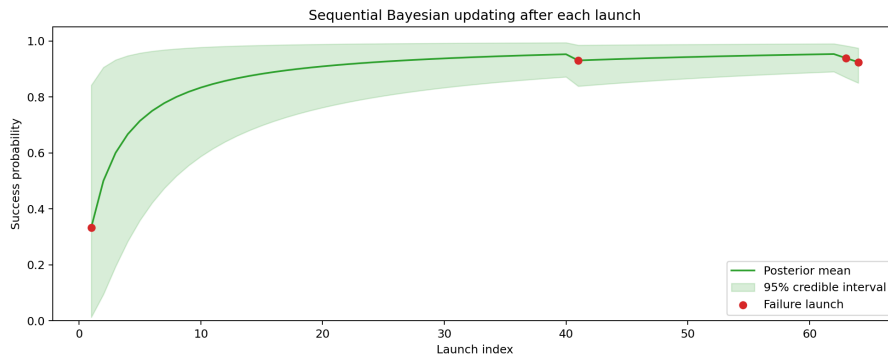
**Fig. 19** Supplementary diagnostic plot from the reproducibility archive (source identifier: step3\_06\_posterior\_predictive\_checks).



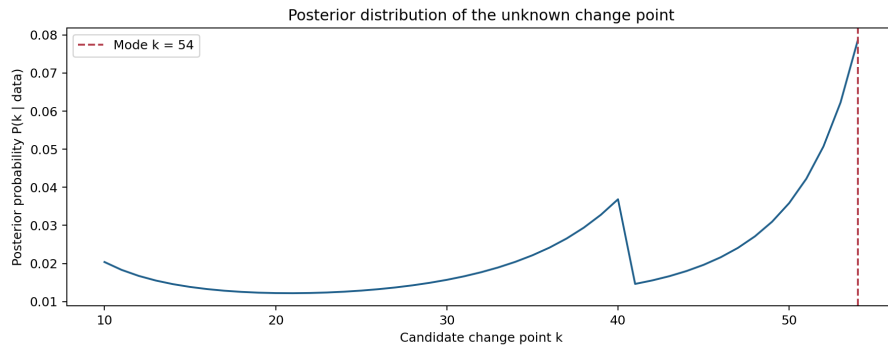
**Fig. 20** Supplementary diagnostic plot from the reproducibility archive (source identifier: step3\_07\_historical\_update).



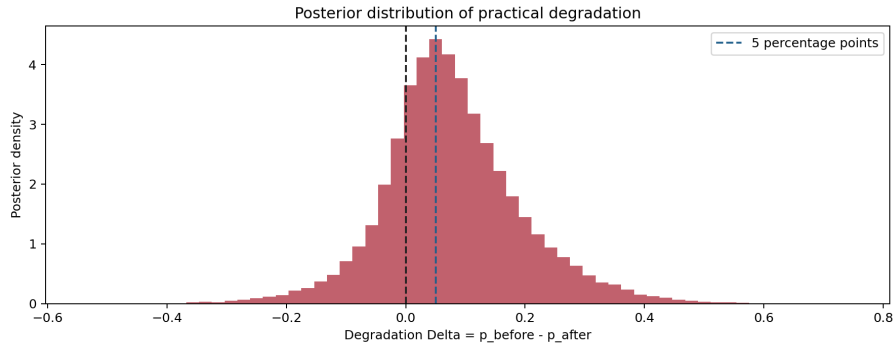
**Fig. 21** Supplementary diagnostic plot from the reproducibility archive (source identifier: step3\_08\_recent\_failure\_sensitivity).



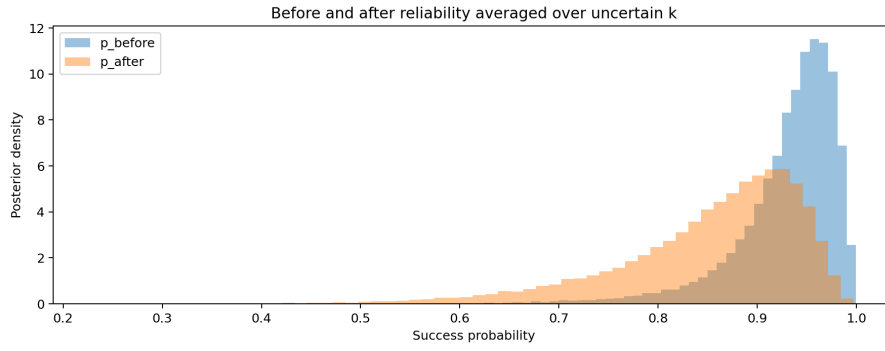
**Fig. 22** Supplementary diagnostic plot from the reproducibility archive (source identifier: step3\_09\_sequential\_bayesian\_update).



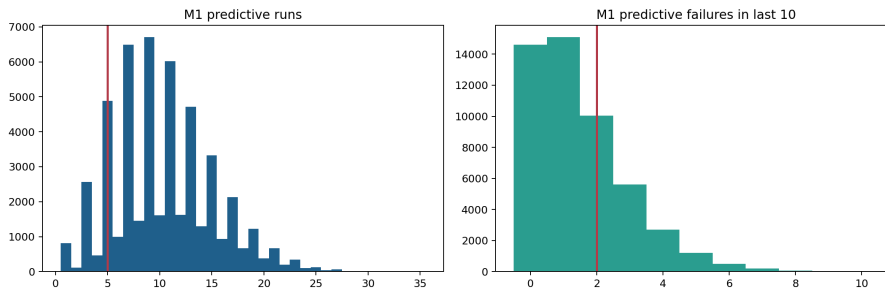
**Fig. 23** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_01\_change\_point\_posterior).



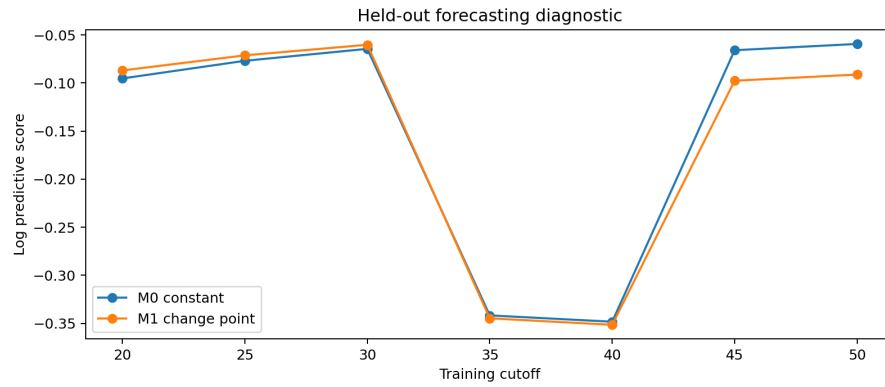
**Fig. 24** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_02\_degradation\_posterior).



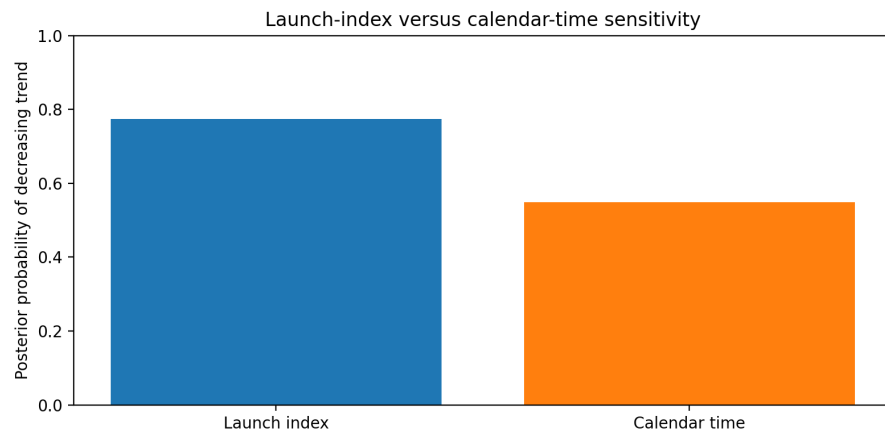
**Fig. 25** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_03\_before\_after\_posteriors).



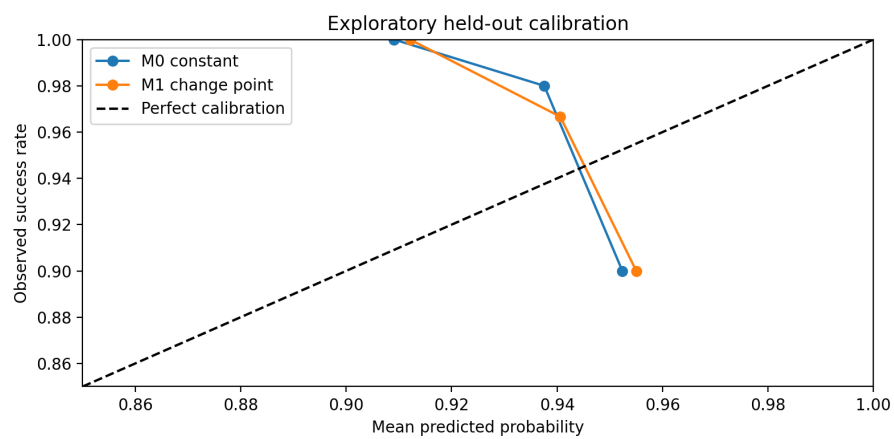
**Fig. 26** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_04\_change\_point\_ppc).



**Fig. 27** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_05\_out\_of\_sample\_scores).



**Fig. 28** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_06\_calendar\_vs\_index\_time).



**Fig. 29** Supplementary diagnostic plot from the reproducibility archive (source identifier: step4\_07\_calibration).

## Declarations

### Funding

No external funding was received for this study.

### Competing interests

The authors declare no competing interests.

### Ethics approval and consent to participate

Not applicable. The study uses historical public launch records and contains no human or animal subjects.

### Data availability

The source-audited dataset, data dictionary, source audit, data-freeze statement, and validation report are included in the accompanying reproducibility package.

### Code availability

The data-construction script, analysis notebooks, and simulation and sequential-validation scripts are included in the reproducibility package. Random seeds, priors, validation windows, and analysis outputs are recorded in a versioned analysis manifest.

### Author contributions

U.R. conceived the study, designed the Bayesian surveillance framework, constructed the data audit, performed the analysis, prepared the figures, and drafted the manuscript. A.G. contributed to the PSLV mission-data interpretation, engineering context, methodological review, and critical revision of the manuscript. Both authors approved the final manuscript.

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